Assignment-9

Principal Component Analysis (PCA) using Hald Cement Dataset

Objective

1. To understand Principal Component Analysis (PCA).
2. To reduce dimensionality of correlated variables.
3. To identify principal components explaining maximum variance.
4. To visualize relationships between variables using PCA plots.

Dataset

Hald Cement Dataset

Variables used in the dataset:

Variable	Description
X1	Tricalcium aluminate
X2	Tricalcium silicate
X3	Tetracalcium aluminoferrite
X4	Dicalcium silicate
Y	Heat evolved during hardening

For PCA we will analyze only the predictor variables **X1, X2, X3, X4**.

```
library(corrplot)
```

```
corrplot 0.95 loaded
```

```
library(factoextra)
```

```
Loading required package: ggplot2

Welcome to factoextra!

Want to learn more? See two factoextra-related books at
https://www.datanovia.com/en/product/practical-guide-to-principal-component-methods-in-r/
```

```
library(GGally)
```

Loading Data Set

```
setwd("D:/Sem-2/R-/7,8,9,10/7")
data<-read.csv("BEST-Hald Cement data.csv")
print(data)
```

```
       Y X1 X2 X3 X4
1   78.5  7 26  6 60
2   74.3  1 29 15 52
3  104.3 11 56  8 20
4   87.6 11 31  8 47
5   95.9  7 52  6 33
6  109.2 11 55  9 22
7  102.7  3 71 17  6
8   72.5  1 31 22 44
9   93.1  2 54 18 22
10 115.9 21 47  4 26
11  83.8  1 40 23 34
12 113.3 11 66  9 12
13 109.4 10 68  8 12
```

```
str(data)
```

```
'data.frame':   13 obs. of  5 variables:
 $ Y : num  78.5 74.3 104.3 87.6 95.9 ...
 $ X1: int  7 1 11 11 7 11 3 1 2 21 ...
 $ X2: int  26 29 56 31 52 55 71 31 54 47 ...
 $ X3: int  6 15 8 8 6 9 17 22 18 4 ...
 $ X4: int  60 52 20 47 33 22 6 44 22 26 ...
```

```
summary(data)
```

```
       Y                X1               X2              X3              X4
 Min.   : 72.50   Min.   : 1.000   Min.   :26.00   Min.   : 4.00   Min.   : 6
 1st Qu.: 83.80   1st Qu.: 2.000   1st Qu.:31.00   1st Qu.: 8.00   1st Qu.:20
 Median : 95.90   Median : 7.000   Median :52.00   Median : 9.00   Median :26
 Mean   : 95.42   Mean   : 7.462   Mean   :48.15   Mean   :11.77   Mean   :30
 3rd Qu.:109.20   3rd Qu.:11.000   3rd Qu.:56.00   3rd Qu.:17.00   3rd Qu.:44
 Max.   :115.90   Max.   :21.000   Max.   :71.00   Max.   :23.00   Max.   :60
```

```
cor(data)
```

```
            Y         X1         X2         X3         X4
Y   1.0000000  0.7307175  0.8162526 -0.5346707 -0.8213050
X1  0.7307175  1.0000000  0.2285795 -0.8241338 -0.2454451
```

```
X2  0.8162526  0.2285795  1.0000000 -0.1392424 -0.9729550
X3 -0.5346707 -0.8241338 -0.1392424  1.0000000  0.0295370
X4 -0.8213050 -0.2454451 -0.9729550  0.0295370  1.0000000
```

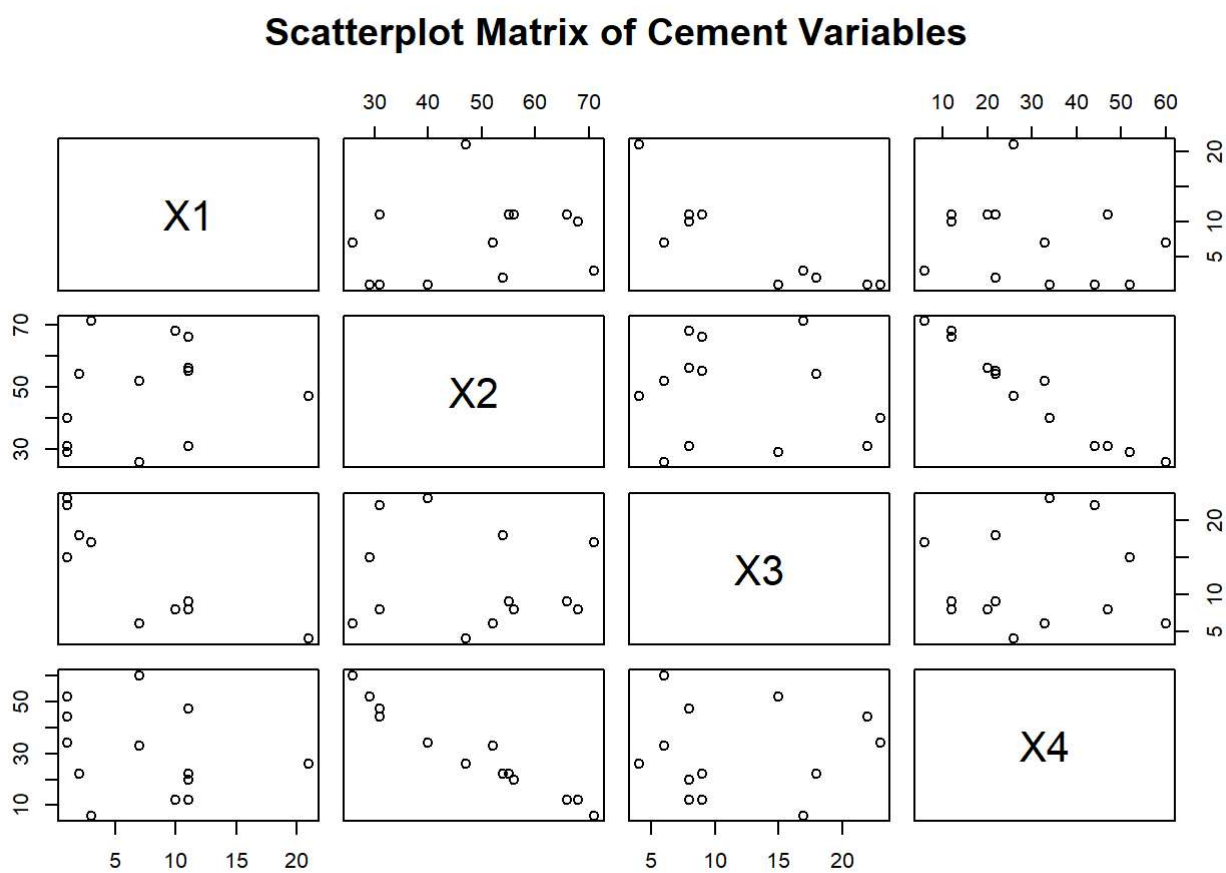
Select Predictor Variable

PCA Should only be applied to predictor Variables

```
predictors <- data[,c("X1","X2","X3","X4")]
```

Scatterplot Matrix

```
pairs(predictors,
main="Scatterplot Matrix of Cement Variables")
```



Correlation Matrix

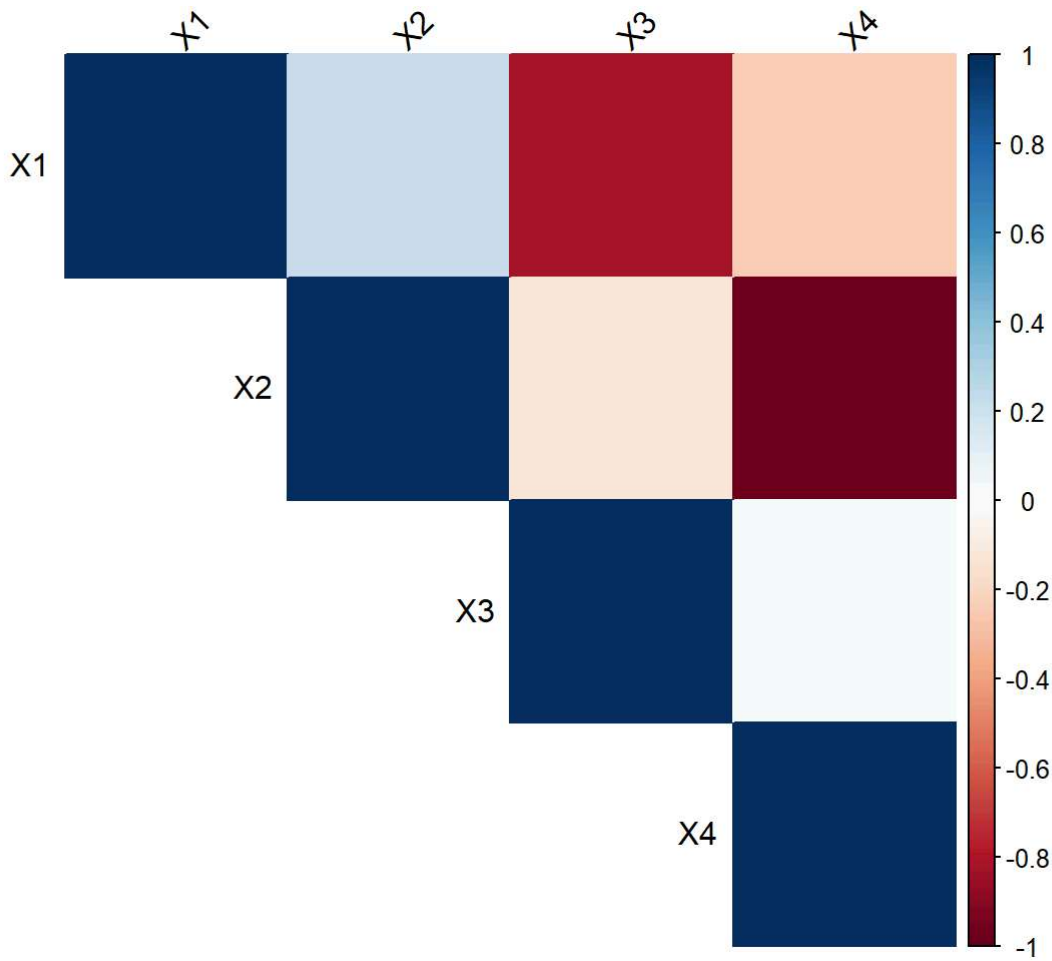
```
cor_matrix <- cor(predictors)

cor_matrix
```

```
           X1         X2         X3         X4
X1  1.0000000  0.2285795 -0.8241338 -0.2454451
X2  0.2285795  1.0000000 -0.1392424 -0.9729550
X3 -0.8241338 -0.1392424  1.0000000  0.0295370
X4 -0.2454451 -0.9729550  0.0295370  1.0000000
```

Correlation Heatmap

```
corrplot(cor_matrix,
method="color",
type="upper",
tl.col="black",
tl.srt=45)
```



Standardize Varialbles

- PCA requires Variables to be standardized

```
scaled_data <- scale(predictors)
```

Doing PCA now

```
pca_model <- prcomp(scaled_data)

summary(pca_model)
```

```
Importance of components:
                          PC1    PC2     PC3     PC4
Standard deviation     1.4952 1.2554 0.43198 0.04030
Proportion of Variance 0.5589 0.3940 0.04665 0.00041
Cumulative Proportion  0.5589 0.9529 0.99959 1.00000
```

- Eigen values here represents the variance explained by each principle component

```
pca_model$sdev^2
```

```
[1] 2.235704035 1.576066070 0.186606149 0.001623746
```

shows how each variable contributes to component

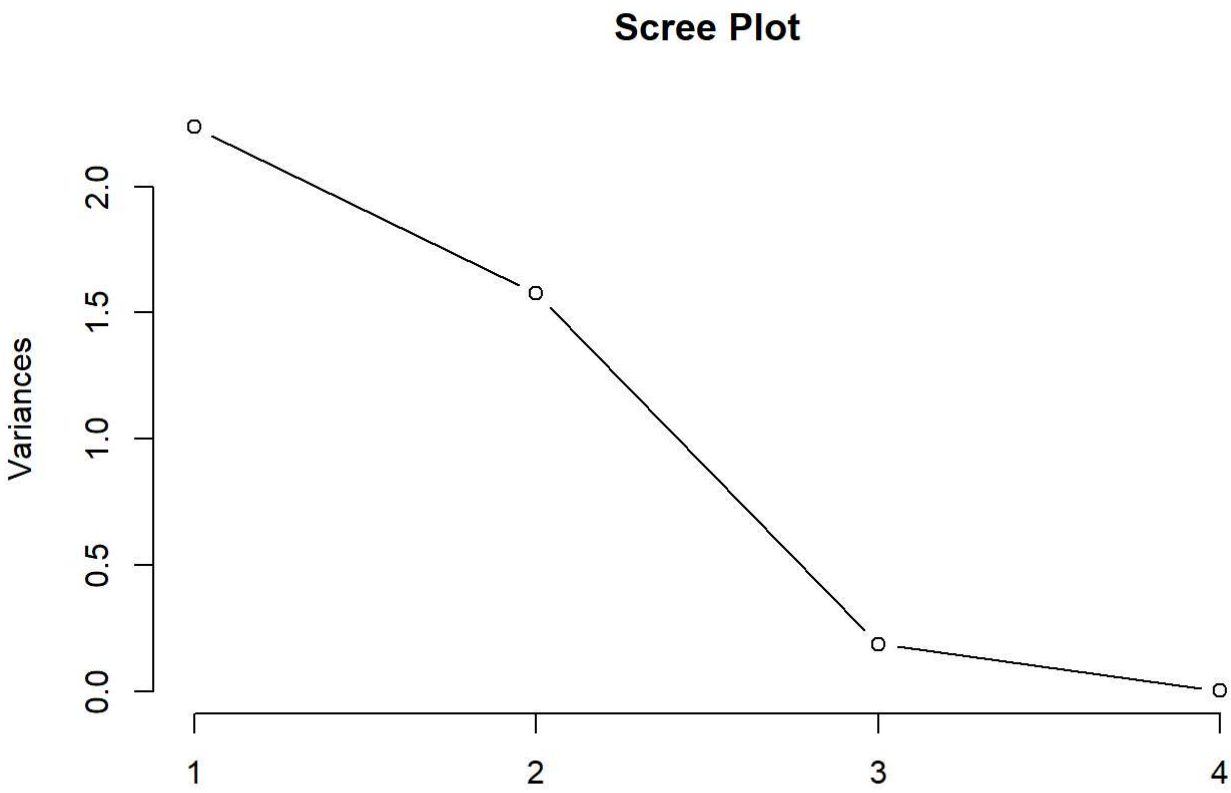
- PCA Loading

```
pca_model$rotation
```

```
          PC1        PC2        PC3       PC4
X1  0.4759552  0.5089794 -0.6755002 0.2410522
X2  0.5638702 -0.4139315  0.3144204 0.6417561
X3 -0.3940665 -0.6049691 -0.6376911 0.2684661
X4 -0.5479312  0.4512351  0.1954210 0.6767340
```

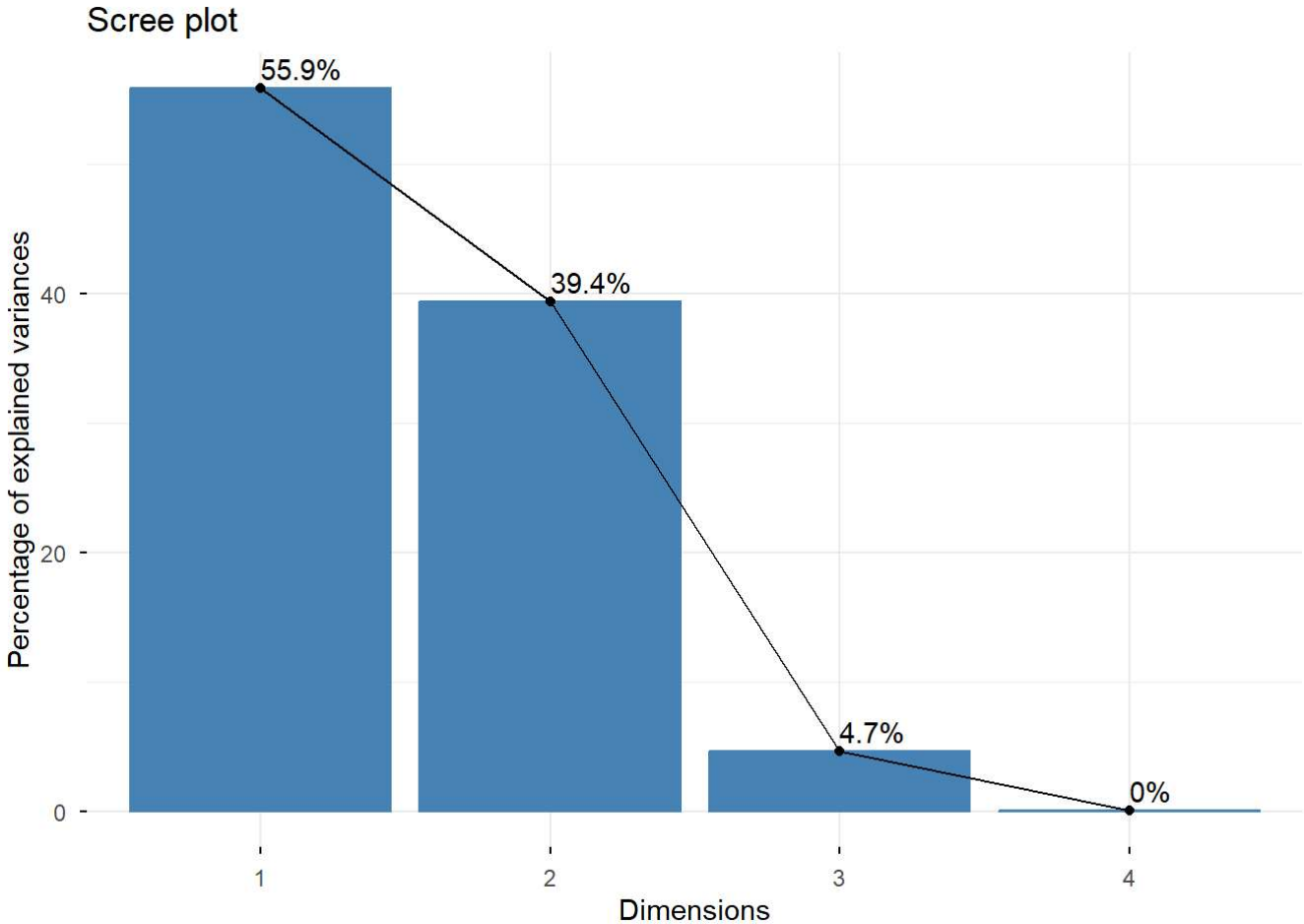
Screen Plot

```
plot(pca_model,
type="l",
main="Scree Plot")
```



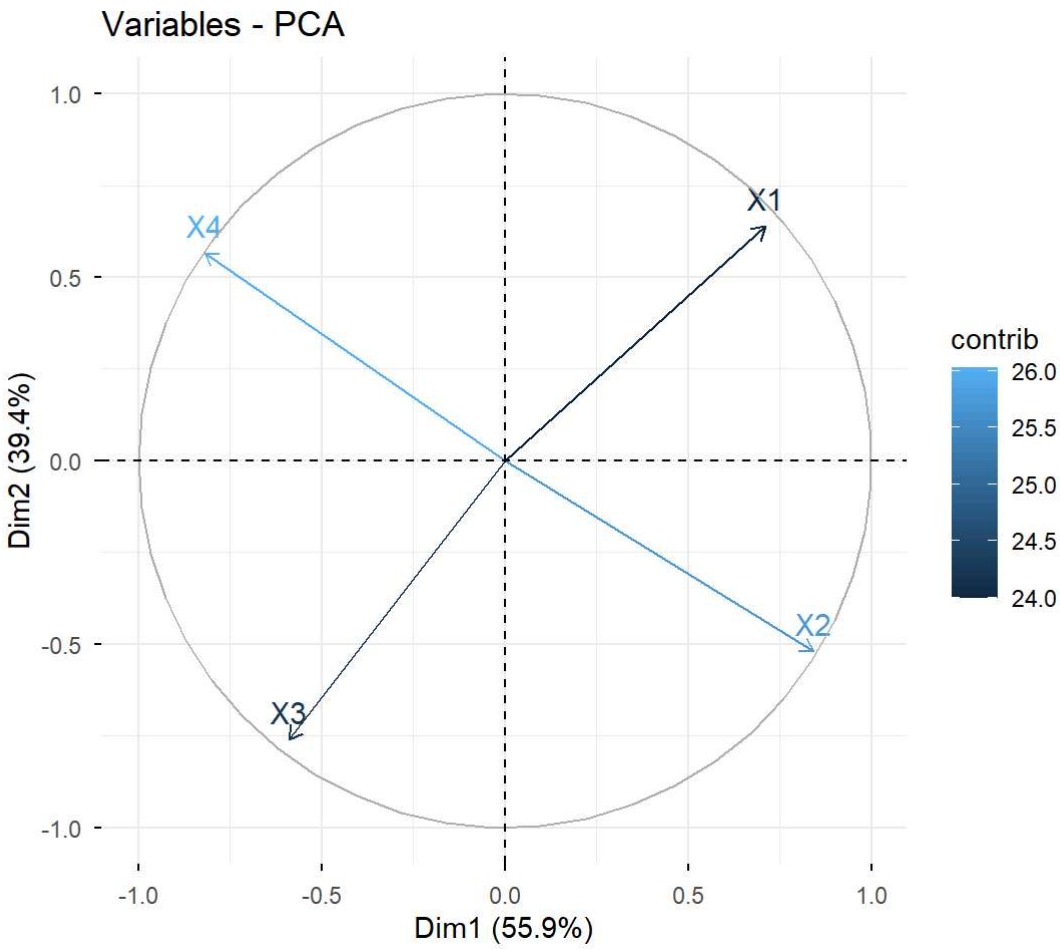
Eigenvalue Visualization

```
fviz_eig(pca_model,
addlabels=TRUE)
```



Variable Contribution PLot

```
fviz_pca_var(pca_model,
col.var="contrib")
```

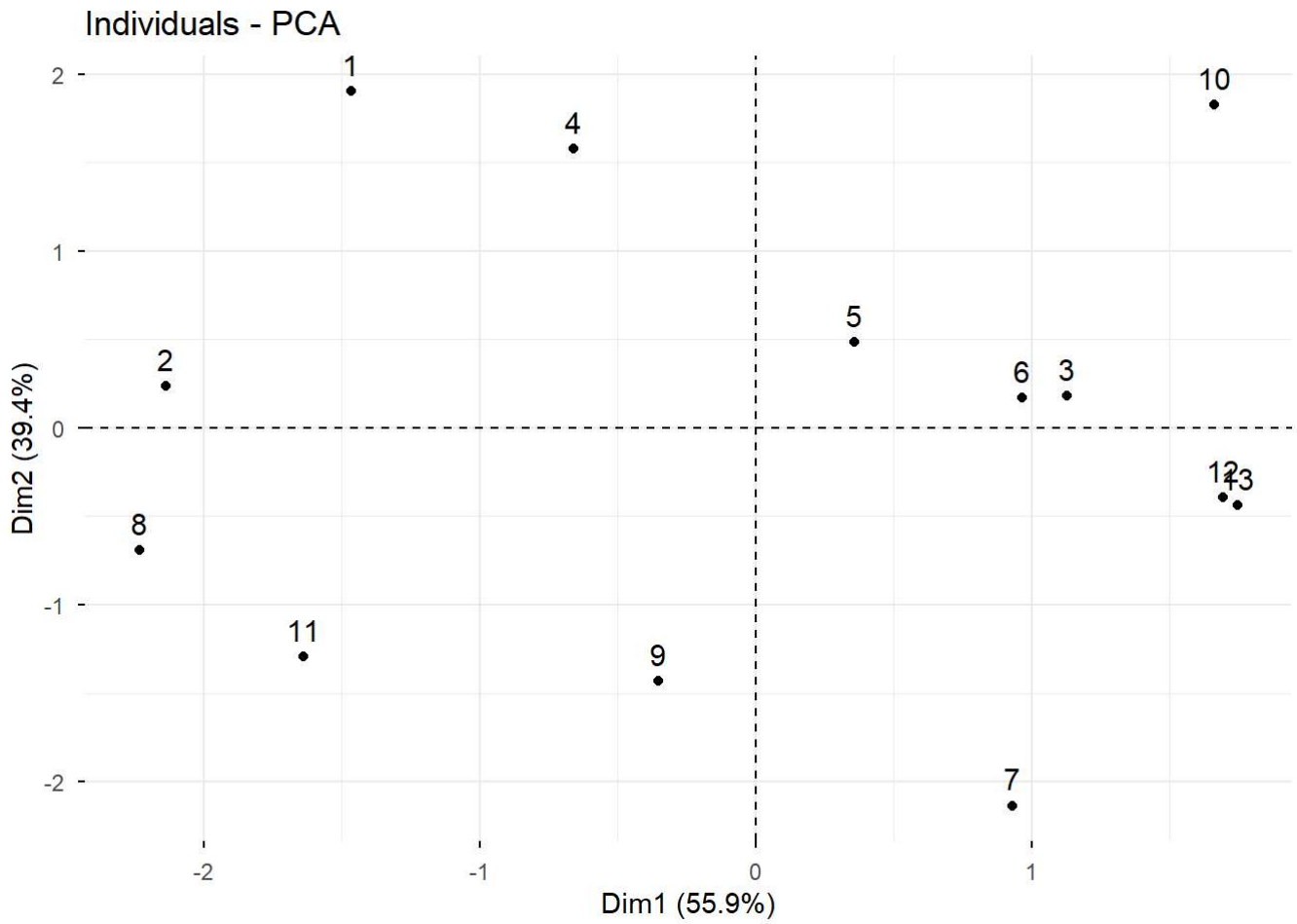



```
summary(pca_model)$importance
```

```
                          PC1      PC2       PC3        PC4
Standard deviation     1.495227 1.255415 0.4319793 0.04029573
Proportion of Variance 0.558930 0.394020 0.0466500 0.00041000
Cumulative Proportion  0.558930 0.952940 0.9995900 1.00000000
```

Observation Plot

```
fviz_pca_ind(pca_model)
```



Extracting Principal Components

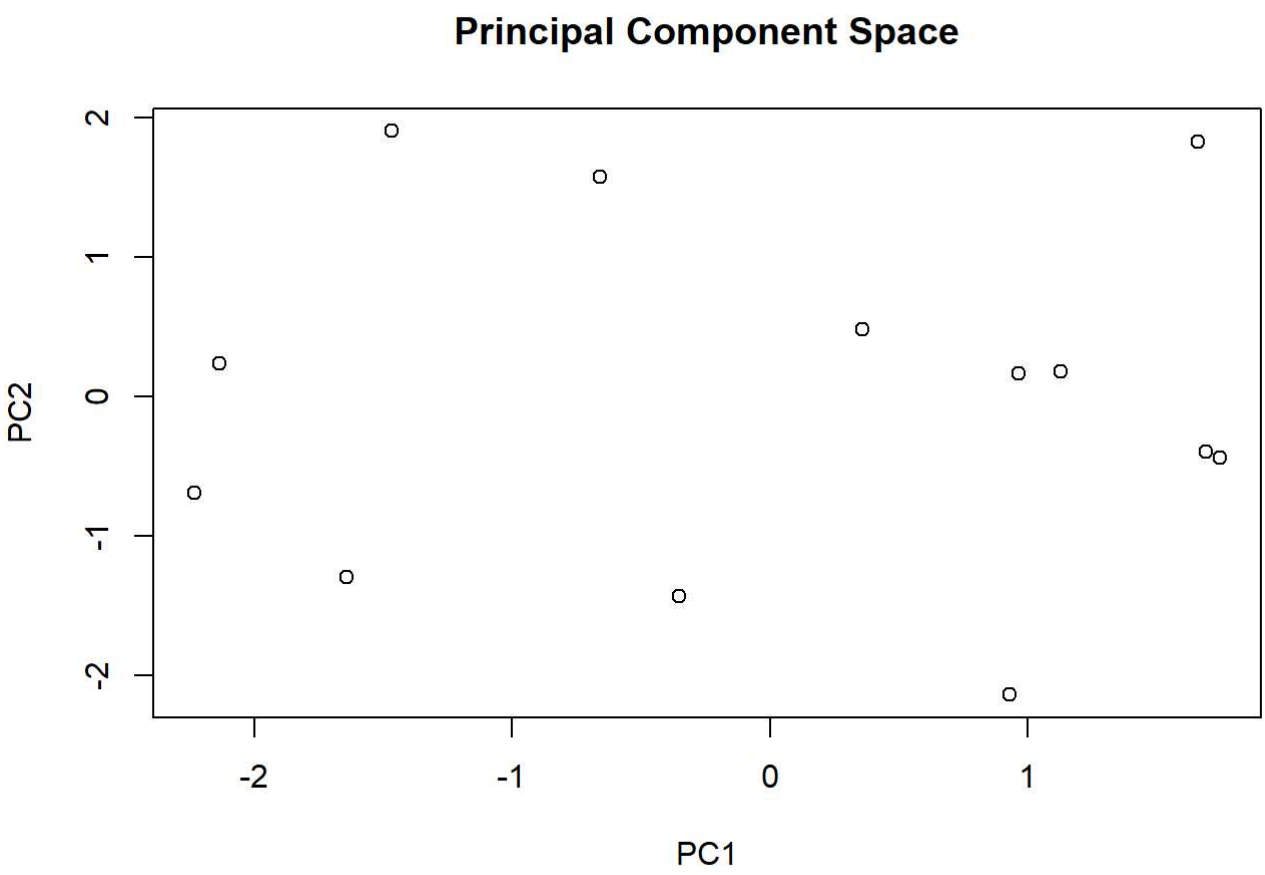
```
pc_scores <- pca_model$x

head(pc_scores)
```

```
            PC1       PC2         PC3         PC4
[1,] -1.4672378 1.9030357  0.53000021  0.03852996
[2,] -2.1358287 0.2383537  0.29018631 -0.02983282
[3,]  1.1298705 0.1838772  0.01071260 -0.09370084
[4,] -0.6598955 1.5767742 -0.17920364 -0.03311636
[5,]  0.3587646 0.4835379  0.74012228  0.01918711
[6,]  0.9666396 0.1699440 -0.08570239 -0.01216708
```

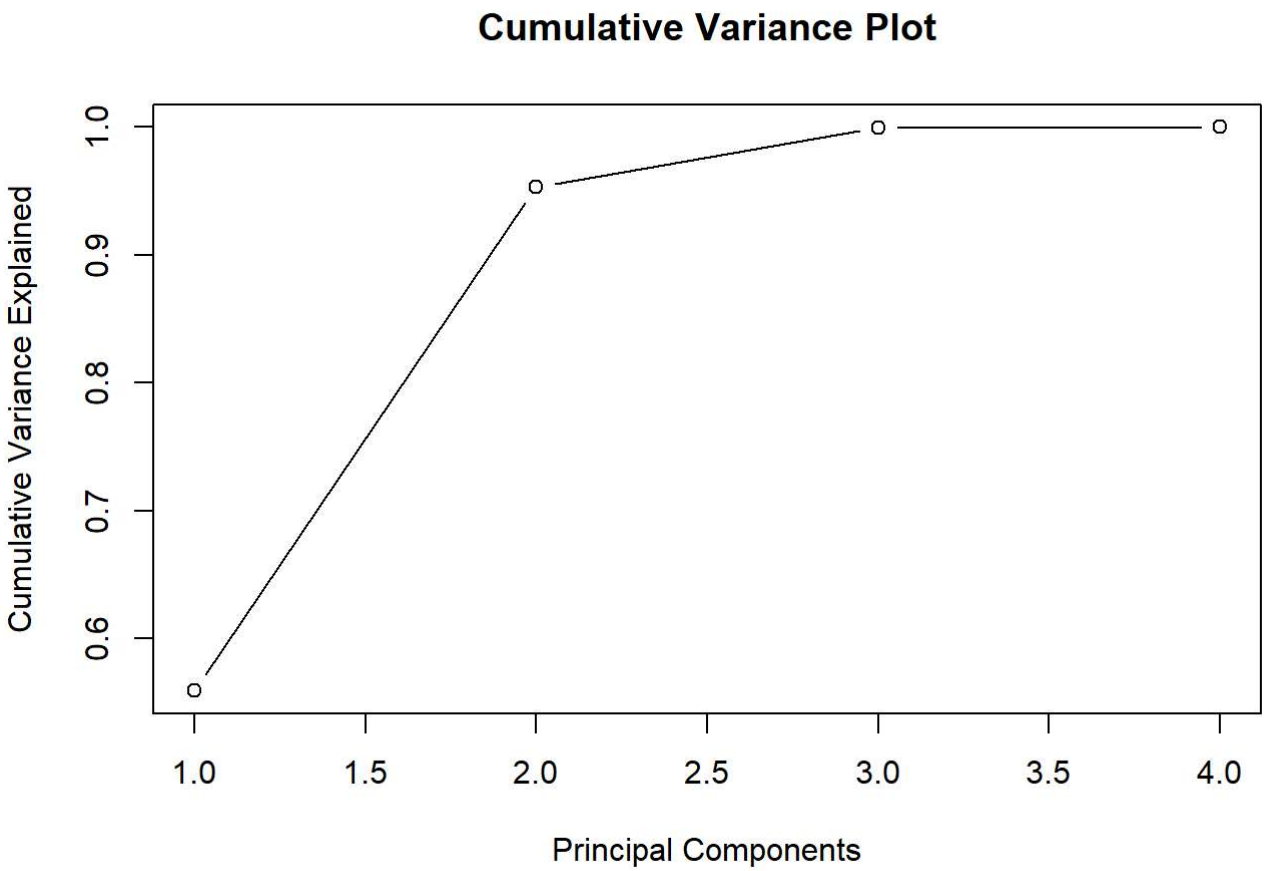
Visualizing first two components

```
plot(pc_scores[,1],
pc_scores[,2],
xlab="PC1",
ylab="PC2",
main="Principal Component Space")
```



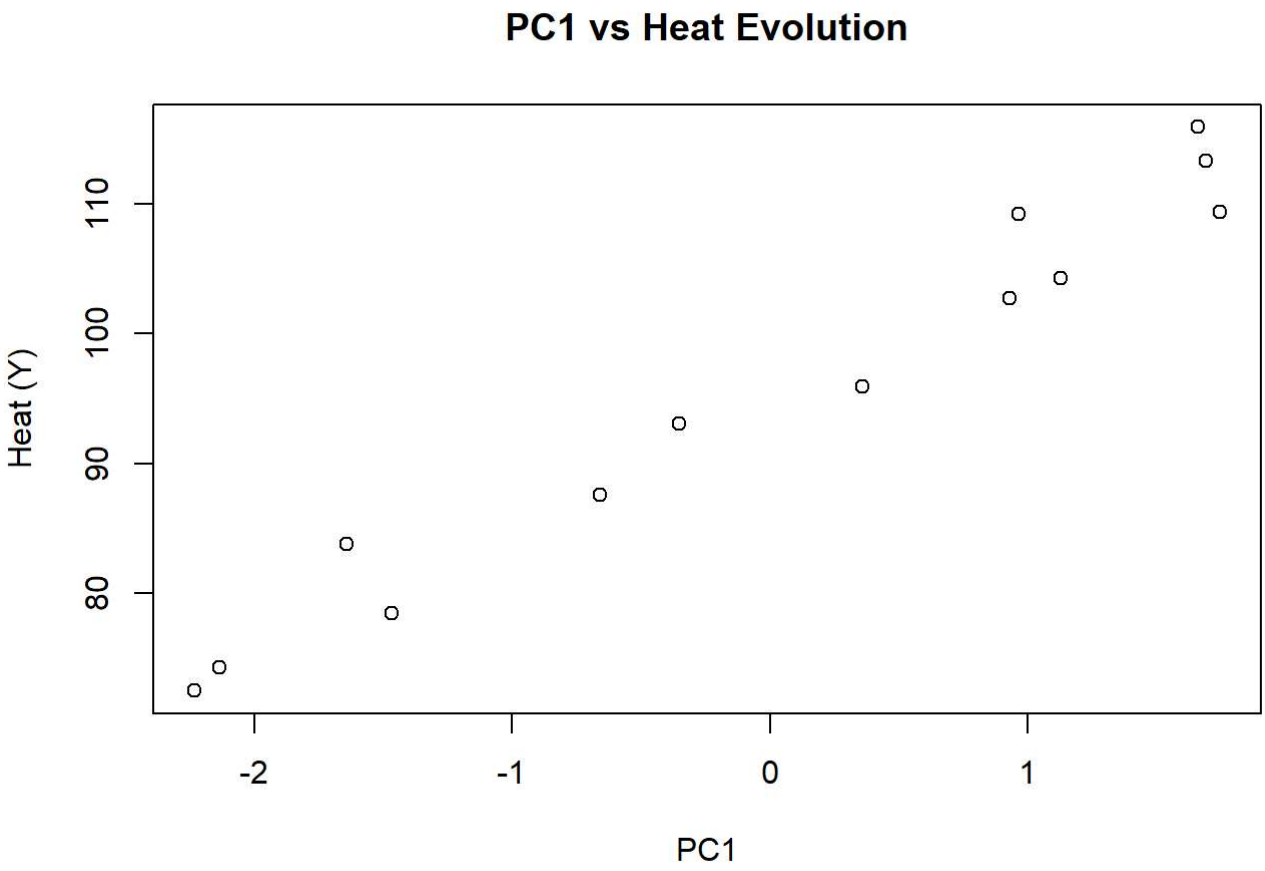

```
var_explained <- pca_model$sdev^2 / sum(pca_model$sdev^2)

plot(cumsum(var_explained),
     type="b",
     xlab="Principal Components",
     ylab="Cumulative Variance Explained",
     main="Cumulative Variance Plot")
```



Relation ship betwwn PC1 and target variable

```
plot(pc_scores[,1], data$Y,
     xlab="PC1",
     ylab="Heat (Y)",
     main="PC1 vs Heat Evolution")
```


PC1 vs Heat Evolution
Heat (Y)
80
90
100
110
-2
-1
0
1
PC1